Ashwin K M

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Papers

- **Learn–Master–Teach Tuning: From Pupil to Professor** *Ashwin KM et al.*
  A Human-Pedagogy Inspired Fine-Tuning Paradigm for LLMs
- **FOCAL: Decoupling Conflicting Objectives in LLM Domain Adaptation via Temporal Orthogonalization** *Ashwin KM et al.*

Education

- **Indian Institute of Science, Bangalore** *TGPA: 8.3*
  B.Tech in Mathematics and Computing 2023 – 2027

Research Experience

- **Quest Lab**
  Research Intern Aug 2026 – Dec 2026
  - Formulated a product vision for a **configurable turbine digital-twin platform**, motivated by the lab's need to connect manufacturers and customers through shared equipment context, organization-specific fleet monitoring, and explicitly scoped access across sites.
  - Implemented **telemetry replay, interactive turbine schematics, and scoped access controls**, integrating authorization and provisioning components around equipment identity, sensor mappings, recoverable setup, and buyer-controlled sharing to support investigations across organizations and sites.
  - Designed an **evidence-linked investigation workflow** combining historical telemetry, operating context, copilot explanations, and cited procedures, with maintenance proposals routed for human review and diagnostic hypotheses kept distinct from measured observations.
- **Zenteiq AI**
  ML Research Engineer May 2025 – Dec 2025
  - Investigated whether training LLMs through a human-inspired **Learn–Master–Teach** lifecycle improves mathematical reasoning accuracy beyond static fine-tuning.
  - Designed a multi-stage training pipeline combining curriculum-grounded learning, interactive concept refinement, feedback-driven practice, multi-teacher distillation, self-reflection, hallucination mitigation, and pedagogical reinforcement learning to systematically improve reasoning reliability.
  - Found that the integrated pipeline yields **+10–17 pp accuracy gains** over the base model on CBSE-style math benchmarks, while revealing that feedback-driven refinement and teaching-oriented objectives are critical for stable gains beyond knowledge injection alone.

Selected Projects

- **Selective Token Memory for Recurrent Language Models** *Aug 2026 – Dec 2026*
  Supervised by Prof. Shirish Shevade
  - Investigated how recurrent language models can preserve and selectively access historical evidence, analyzing how fixed-state compression, segment-level summaries, and averaging of retrieved memories in Hierarchical Memory Mamba (HMM) can obscure distinctions needed for later predictions, and motivating a separation between locating relevant history and reading its detailed contents.
  - Developed a **selective token-memory adapter** for frozen Mamba2 that combines a dedicated sliding-window path with selective attention over encoded tokens in distant chunks. Adapted RAMba's hierarchical sparse attention to separate coarse chunk selection from fine-grained reads, and used selected relevance scores in differentiable fusion weights to connect memory selection to prediction loss.
  - Evaluated the complete adapter against local HMM on Mamba2-1.3B under a shared PG-19 protocol across three paired seeds, reducing mean 32K perplexity from 13.384 to 13.006 (**2.82%**) with improvements at all four evaluated context lengths from 4K to 32K, demonstrating stronger long-context language modeling while retaining the pretrained recurrent backbone.
- **FOCAL** *Jan 2026 – Apr 2026*
  Supervised by Prof. Shalabh Bhatnagar
  - Current AI scientists fail to identify meaningful research directions for scientific discovery in domains with limited prior work as they don't build foundational understanding nor iteratively refine their hypothesis.
  - Proposed a multi-agent framework that acquires prerequisite knowledge, generates and critiques research hypotheses, and designs experiments to evaluate promising research directions.
  - Demonstrated up to 13.2 percentage-point improvements over ABMLL across common-sense, mathematical, coding, and medical reasoning benchmarks, while exhibiting stronger out-of-domain generalization than competing meta-learning approaches.

- **Latent Diffusion Research Agents** *Jan 2026 – May 2026*
  Supervised by Prof. Y. Narahari
  - Current deep research agents rely on repeated text-mediated communication between agents, introducing cumulative information loss, vocabulary bottlenecks, and increasing inference costs across long-horizon research workflows.
  - Developed a latent-space collaboration framework that replaces internal text exchanges with KV-cache transfer, localizes decoding to tool-interaction boundaries, and provides theoretical guarantees on convergence, information preservation, and iterative agent refinement.
  - Matched the performance of text-mediated baselines while reducing generated tokens by 33% and wall-clock execution time by 28%, validating latent communication as a more efficient mechanism for multi-agent coordination.

- **Benchmarking Offline Reinforcement Learning Algorithms** *Jan 2025 – Apr 2025*
  Supervised by Prof. Chiranjib Bhattacharya
  - Investigated the robustness of standard offline reinforcement learning algorithms (Decision Transformer, Implicit Q-Learning, TD3-BC, AWAC, ReBRAC, Behavior Cloning) by benchmarking them on MuJoCo continuous-control tasks using the **Minari dataset format**, motivated by the need to move beyond legacy D4RL pipelines and evaluate algorithms under modern, reproducible offline-RL data interfaces.
  - Refactored and rebuilt the offline RL training and evaluation pipeline to support **Minari + MuJoCo environments**, addressing incompatibilities with D4RL assumptions (dataset APIs, environment registration, trajectory handling), and ensuring fair, like-for-like comparisons across algorithms under a unified experimental setup.
  - Conducted an architectural ablation replacing standard `nn.Linear` layers with `NdLinear` layers across policies and critics, observing marginal performance differences in single-dimensional settings but identifying a key limitation of the benchmarknamely that NdLinears advantages are likely to emerge in higher-dimensional tensor structureshighlighting a concrete direction for future multi-modal or structured-state offline RL evaluation.

Skills

- **Programming & Core ML:** Python, PyTorch, HuggingFace, LLM Routing, Multi-Fidelity Systems
- **LLMs & Agentic Systems:** PEFT, LangChain, LangGraph, vLLM, Unsloth
- **Reinforcement Learning:** Multi-Armed Bandits, TRL, OpenRLHF, Gymnasium, MuJoCo
- **Systems & Infrastructure:** Efficient Inference, AWS Cloud, Ray, Docker, Git, Weights & Biases

Achievements

- **World Scholars Cup, Global Round (Manila)** - Placed **30th out of 1,000 teams** worldwide; won **8 Gold and 5 Silver medals**; qualified for the **Tournament of Champions at Yale University** *2019*
- **National Talent Search Examination (NTSE)** - Stage I Awardee with **Rank 55** *2021*
- **Kishore Vaigyanik Protsahan Yojana (KVPY)** - Secured **Top 2000 rank** in SA stream *2021*

Positions of Responsibility

- **Core Coordinator Nexus (SciTech Fest), Rhapsody 2025, IISc** *Dec 2024 – Mar 2025*
  - Founded and led **Nexus**, scaling it into a flagship SciTech platform at IISc focused on **AI, Robotics, IoT, and Automation**.
  - Drove **industry partnerships and large-scale competitions**, collaborating with **Autodesk, L&T, OLA, ART-PARK** and spearheading **Robonautica, Code Conquest, InnovateX**.

- **Sponsorship Coordinator Pravega XI, IISc** *Aug 2024 – Dec 2024*
  - Led **sponsorship acquisition and corporate outreach**, securing substantial funding and long-term industry partnerships.
  - Aligned **corporate branding with Pravegas science and technology mission**, strengthening sponsor retention and studentindustry collaboration.

- **Event Coordinator RevCoding, Pravega 2025, IISc** *Jul 2024 – Jan 2025*
  - Conceptualized and organized **RevCoding**, a reverse-coding competition integrating **algorithmic reasoning, logical deduction, and narrative-driven problem solving**.
  - Designed multi-stage challenges requiring participants to decode hidden clues across code-based problem settings.